

# Enhancing Crowd Accuracy in AI-Assisted Judgmental Forecasting: Theory and Experimental Evidence

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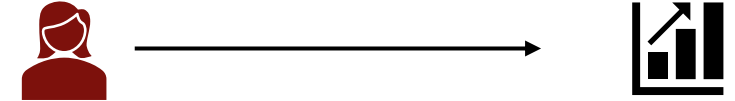


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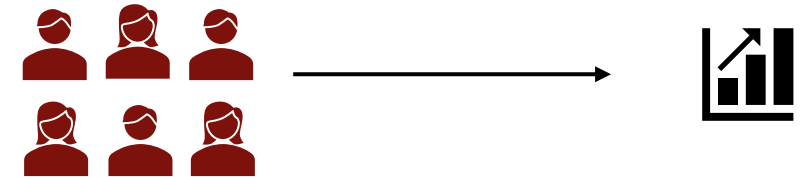
# Humans, Algorithms, and the Wisdom of the Crowd



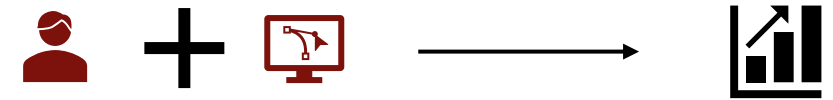
- **Accurate demand forecasts** are important for managerial decisions.



- “**Wisdom of the crowd**” eliminates randomness in human forecasts.



- **Algorithmic advice** can add value to human forecasters.



- **Wisdom of the AI-Assisted Crowd.**





# A Tale of Two Biases

**Borg Effect**

**More AI Advice.**

**Algorithm Aversion**

**Less AI Advice.**

# “Borg” Effect

- Shared-information bias in aggregate forecast (Palley and Soll 2019, Fügener et al. 2021).
- *Individually*, humans are noisy and inaccurate.
  - More reliant on AI advice.
- *In aggregate*, humans are less noisy.
  - Shared use of AI creates a bias.

The crowd average contains **too much AI advice** due to **borg effect**.



## Harnessing the Wisdom of Crowds

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**Abstract.** When will a large group provide an accurate answer to a question involving quantity estimation? We empirically examine this question on a crowd-based corporate earnings forecast platform (Estimize.com). By tracking user activities, we monitor the amount of public information a user views before making an earnings forecast. We find that the more public information users view, the less weight they put on their own private information. Although this improves the accuracy of individual forecasts, it reduces the accuracy of the group consensus forecast because useful private information is prevented from entering the consensus. To address endogeneity concerns related to a user's information acquisition choice, we collaborate with Estimize.com to run experiments that restrict the information available to randomly selected stocks and users. The experiments confirm that “independent” forecasts result in a more accurate consensus. Estimize.com was convinced to switch to a “blind” platform from November 2015 on. The findings suggest that the wisdom of crowds can be better harnessed by encouraging independent voices from among group members and that more public information disclosure may not always improve group decision making.

**History:** Accepted by Renee Adams, finance.

**Supplemental Material:** The online appendix is available at <https://doi.org/10.1287/mnsc.2019.3294>.

**Keywords:** wisdom of crowds • herding • naïve learning • social learning • group decision making • earnings forecast

Da and Huang (2020)

| RESEARCH ARTICLE | COMPUTER SCIENCE



## Generative AI enhances individual creativity but reduces the collective diversity of novel content

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Doshi and Hauser (2024)



# Algorithm Aversion

- Humans can be **algorithm averse** (Dietvorst et al. 2015).
- Behaviorally biased humans **underweight AI advice**.
- This bias propagates into the **aggregate forecast**.

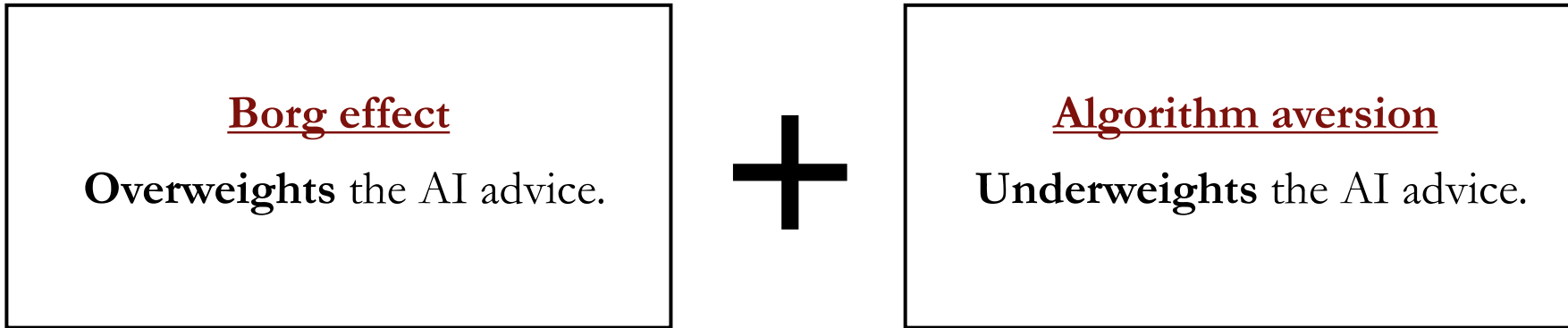
## Even When Algorithms Outperform Humans, People Often Reject Them

By [Jeff Cockrell](#)   October 27, 2020   [CBR - Behavioral Science](#)   [Share This Page](#)

Dietvorst et al. (2015), Chicago Booth Review

The crowd average contains **too little AI advice** due to **algorithm aversion**.

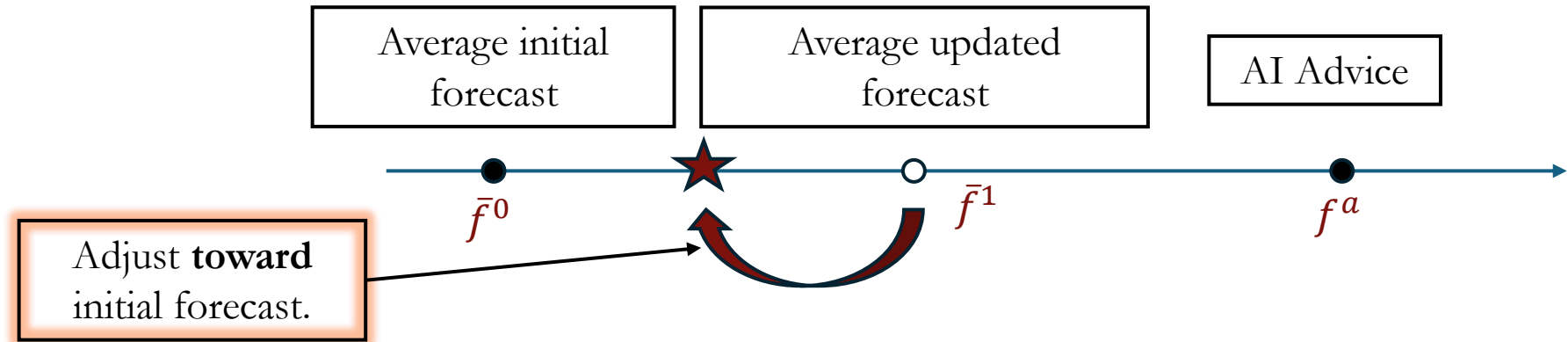
# Biased Crowd Forecasts



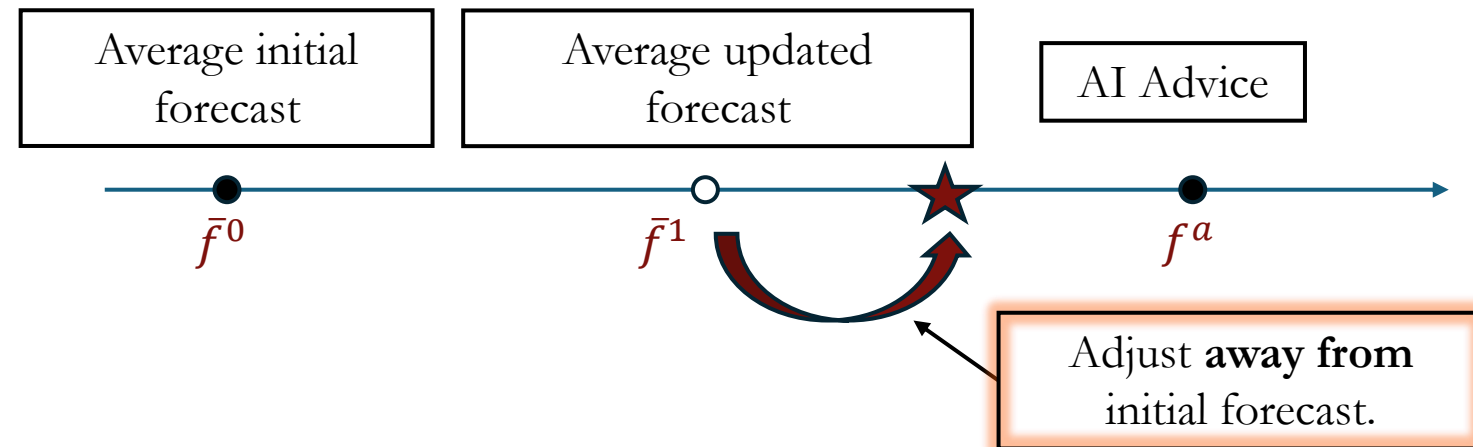
- The net effect on the AI-assisted crowd average can be in **either direction**.
- We can **recalibrate the AI-assisted crowd** to remove the net bias in its forecast.

# Recalibrating the AI-Assisted Crowd

**Borg Effect dominates  
Algorithm Aversion.**



**Algorithm Aversion  
dominates Borg Effect.**



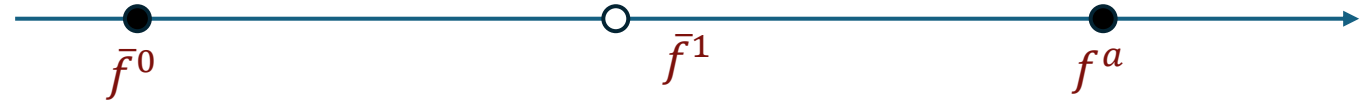
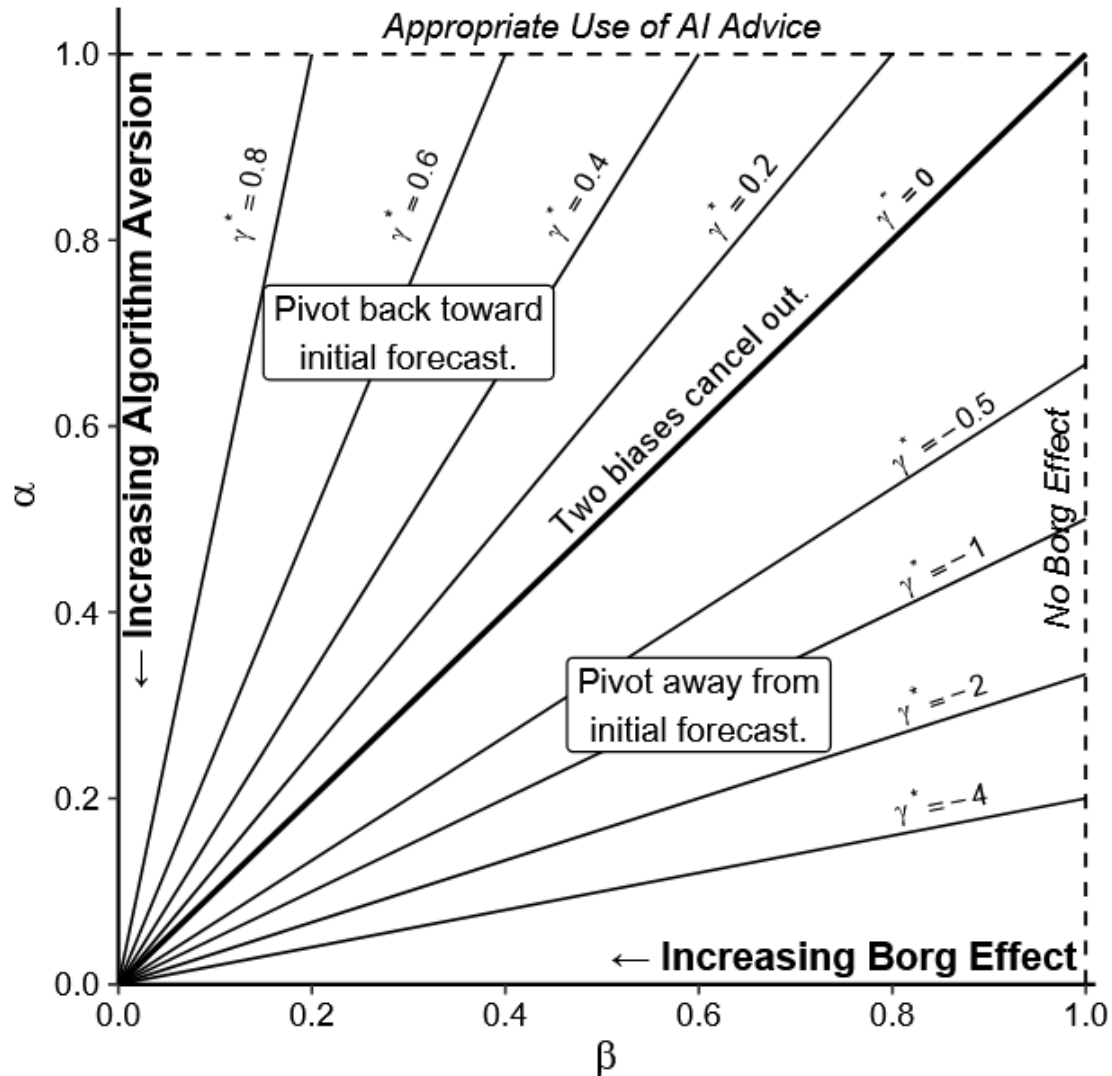
# Optimal Pivoting



Average initial  
forecast

Average updated  
forecast

AI Advice



Optimal pivoting factor

$$\theta^* = \bar{f}^1 + \gamma^*(\bar{f}^0 - \bar{f}^1)$$

We characterize:

$\beta$ :  $\beta_{\text{org}}$  Effect

$\alpha$ : algorithm Aversion

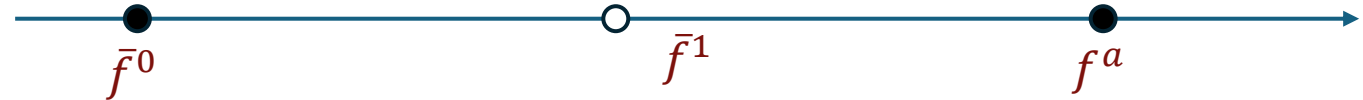
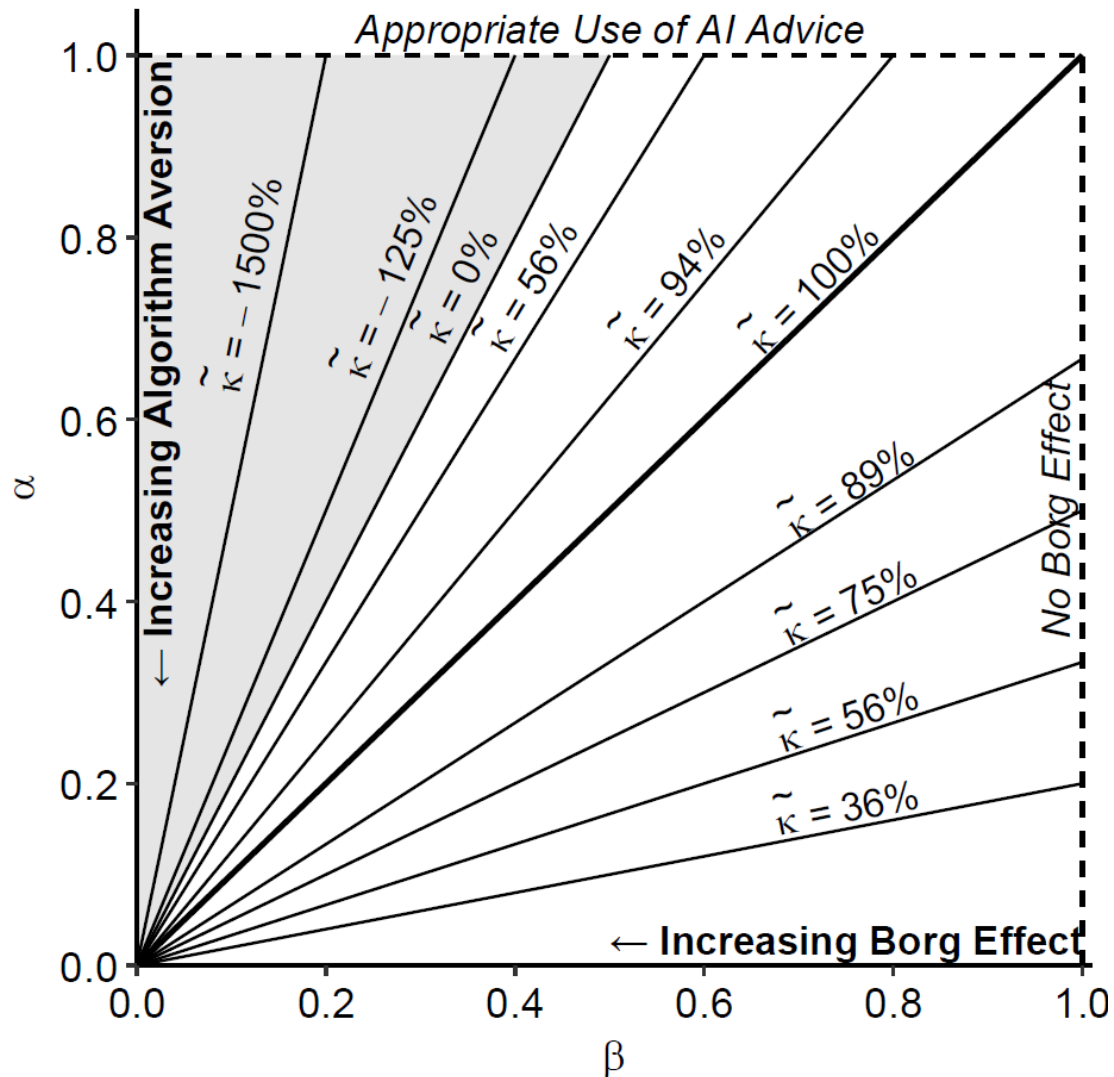
$$\gamma^* = 1 - \frac{\beta}{\alpha}$$

# Optimal Pivoting

Average initial  
forecast

Average updated  
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AI Advice



Optimal pivoting factor

$$\theta^* = \bar{f}^1 + \gamma^*(\bar{f}^0 - \bar{f}^1)$$

We characterize:

$\beta$ :  $\beta_{org}$  Effect

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$$\gamma^* = 1 - \frac{\beta}{\alpha}$$

# Data-Driven Pivoting

- Given historical forecasting data,  $\hat{\gamma}$  minimizes the empirical risk of the pivoted forecast.

$$\hat{\gamma} = \operatorname{argmin}_{\gamma} \sum_{i=1}^N \left( (\bar{f}_i^1 - \theta_i) - \gamma(\bar{f}_i^1 - \bar{f}_i^0) \right)^2$$

- To avoid overfitting due to small training samples, we estimate  $\hat{\gamma}_{\lambda}$  as a LASSO-regularized pivot.

$$\hat{\gamma}_{\lambda} = \operatorname{argmin}_{\gamma} \sum_{i=1}^N \left( (\bar{f}_i^1 - \theta_i) - \gamma(\bar{f}_i^1 - \bar{f}_i^0) \right)^2 + \lambda |\gamma|$$

# Statistical Properties of the Pivoting Factors

- **Asymptotic convergence** of estimated pivoting factors and out-of-sample loss.
- **Controlled loss** in forecasting performance due to regularization penalty.

$$|\hat{\gamma} - \gamma^*| = \mathcal{O}(N^{-1/2})$$

$$L(\hat{\gamma}) - L(\gamma^*) = \mathcal{O}(N^{-1})$$

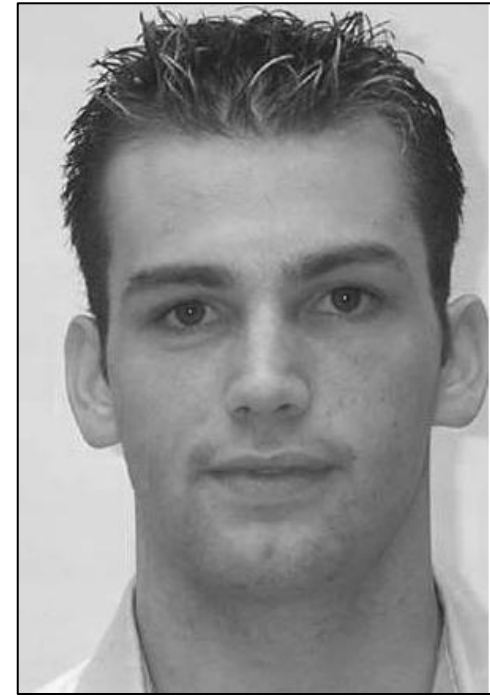
$$|\hat{\gamma}_\lambda - \gamma^*| = \mathcal{O}(\lambda)$$

$$L(\hat{\gamma}_\lambda) - L(\gamma^*) = \mathcal{O}(\lambda^2)$$

# Online Experiments



- Online experiments on Prolific.
- **Make quantitative forecasts using AI advice.**
- Experiment task: Predict ages of people in photographs.



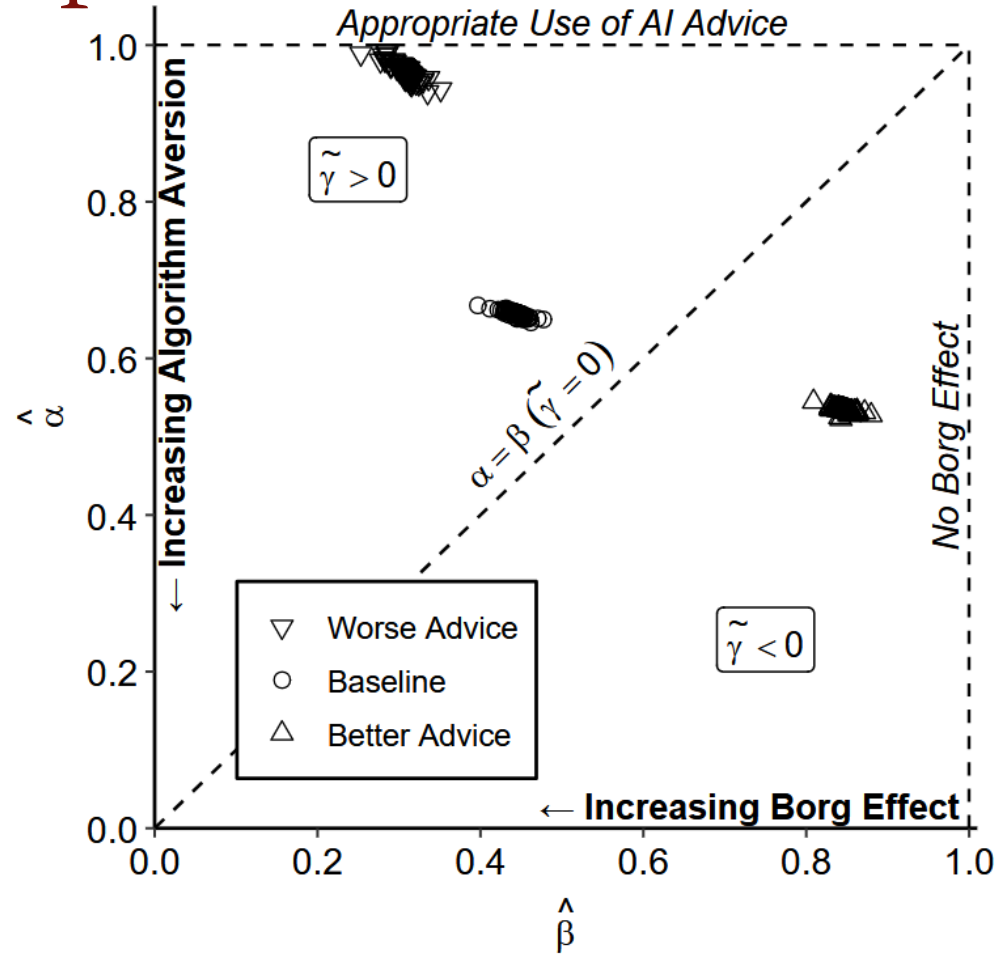
- **Forecasting Environments:**
  - i. Baseline.
  - ii. Worse Advice.
  - iii. Better Advice.

Our AI system estimated that the person in the photograph is **23 years old**, offering the following explanation for its prediction:

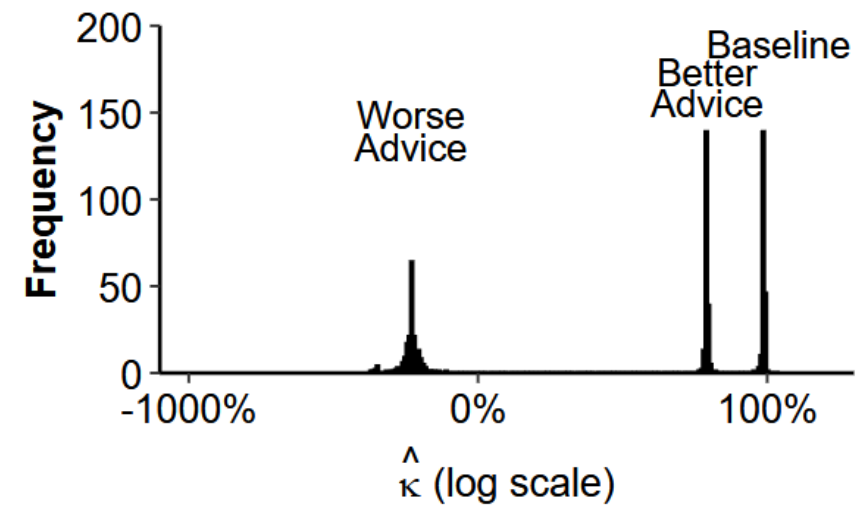
*This age estimate is based on the person's clear skin, youthful features, and the absence of visible wrinkles or other significant signs of aging, which are typically seen in individuals within this age range. The hair is also full and styled in a way that is popular among younger adults.*

In consideration of this advice, what is your best guess of the age of the person in the above photograph?

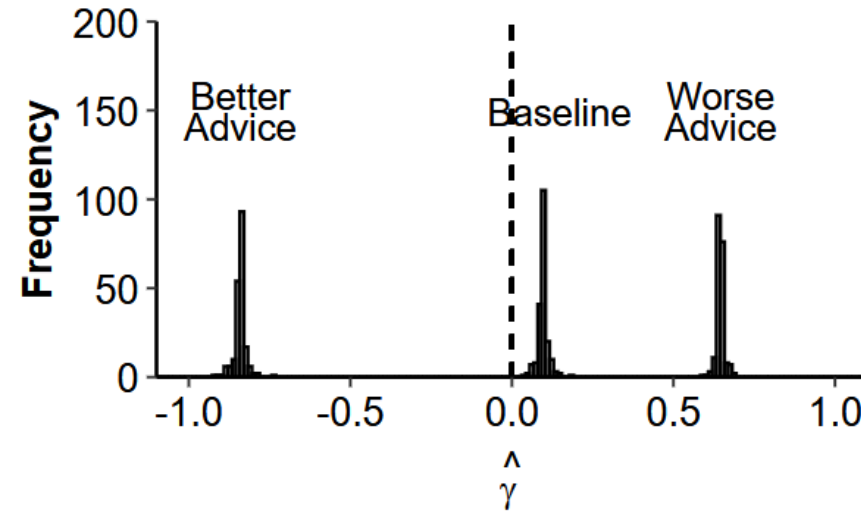
# Experimental Results



(a) Estimated values of borg effect and algorithm aversion.



(b) Estimated values of aggregate AI usage efficiency.



(c) Estimated values of the empirical pivoting factor.

# Comparing the Forecast Accuracies

- We evaluate the **forecasting accuracy** using the regularized pivoting factor.
- Numbers in **bold** denote the *lowest*  $RMSE(s)$  within each treatment.

|                                  | Worse Advice | Baseline    | Better Advice |
|----------------------------------|--------------|-------------|---------------|
| Average $\gamma_\lambda$         | 0.641        | 0.000       | −0.829        |
| RMSE of Average Initial Forecast | 6.42         | 6.54*       | 6.46***       |
| RMSE of AI Advice                | 11.34***     | 7.07***     | 5.64          |
| RMSE of Average Final Forecast   | 6.68*        | <b>6.28</b> | 5.71          |
| RMSE of Pivoted Forecast         | <b>6.36</b>  | <b>6.28</b> | <b>5.54</b>   |

\*: 10%; \*\*: 5%; \*\*\*: 1%.

$$\hat{\theta}_\lambda = \bar{f}^1 + \hat{\gamma}_\lambda(\bar{f}^0 - \bar{f}^1)$$

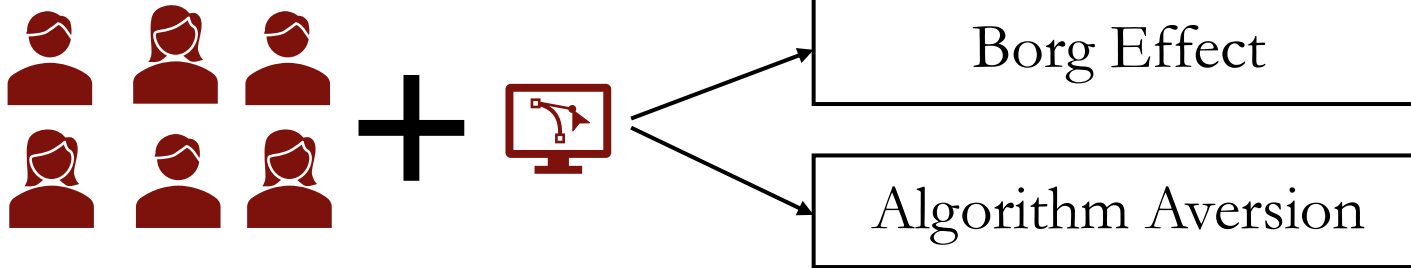
# Pivoting Domains

|                  | Low Algorithm Aversion                                    | High Algorithm Aversion                                 |
|------------------|-----------------------------------------------------------|---------------------------------------------------------|
| Low Borg Effect  | $\gamma = 0$<br>(Stick with the updated forecast.) ✓      | $\gamma < 0$<br>(Pivot away from the initial forecast.) |
| High Borg Effect | $\gamma > 0$<br>(Pivot back toward the initial forecast.) | $\gamma = 0$<br>(Stick with the updated forecast.) ✓    |

- When either bias dominates the other, **pivot appropriately**.
- When both biases are relatively equal in strength, **no need for readjustment**.

# Key Takeaways

The **Wisdom of AI-Assisted Crowds** can be systematically biased.



- 1 **Characterization of biases** in AI-assisted crowd forecasts.
- 2 **Optimal Pivoting** method of recalibrating the consensus forecast.
- 3 Data-driven estimation of the **optimal pivoting factor**.

Paper



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